Responsible AI Governance for Clinician Trust and Adoption of AI-Assisted ASD-EEG Screening

A Mixed-Methods PLS-SEM Investigation of Governance Maturity, Perceived Explainability, Clinician Trust, and Operational Adoption Readiness

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Central Question: *How does Responsible AI governance influence clinician trust and adoption*

The Problem

- Healthcare organizations face significant barriers in **trusting** and **operationally adopting** AI-assisted diagnostic systems
- The barrier is *not* accuracy — classifiers exceed 95%
- The barrier is the absence of:
  - operationalized Responsible AI governance
  - clinician-centered explainability
  - human oversight integrated into workflow
- Result: technically capable AI cannot deliver business value because it cannot pass the trust threshold required for adoption

Research Gap

Existing Research Focus	Missing Area
AI classification accuracy	Clinician trust
EEG prediction methods	Operational governance
Explainability methods (algorithmic)	Operational adoption
AI model performance	Human-centered governance

The dissertation's gap claim

Existing healthcare AI studies focus primarily on predictive performance; insufficient attention has been given to *Responsible AI operationalization* as the antecedent of clinician trust and operational adoption.

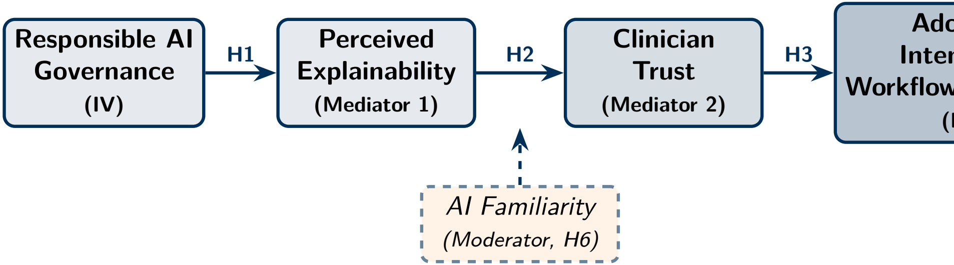
Central Research Question

Locked research question

How does Responsible AI governance influence clinician trust and adoption intention for AI-assisted ASD EEG screening, through perceived explainability?

- One bounded organizational research problem
- Four constructs (Governance, Explainability, Trust, Adoption)
- One mediator chain
- Falsifiable hypotheses

Conceptual Framework



H4: Explainability mediates Governance→Trust; H5: Trust mediates Explainability→Adoption; H6: AI Familiarity mode

- Theoretical anchors: TAM (Davis 1989), Trust in Automation (Lee & See 2004), Socio-Technical Systems (Cherns 1976)

Variables

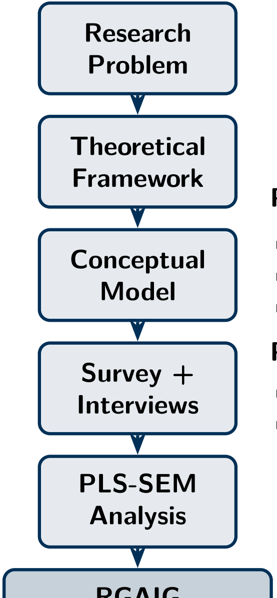
Role	Variable	Measurement
IV	Responsible AI Governance Maturity	12-item scale; 7 control families
Mediator 1	Perceived Explainability	6-item Likert (incl. transparency)
Mediator 2	Clinician Trust	Madsen & Gregor TUI (10 items)
DV	Adoption Intention / Workflow Readiness	TAM-BI + workflow fit (7 items)
Moderator	AI Familiarity	5-point self-rated
Controls	Years of practice, healthcare role, ASD exposure	Demographics

Hypotheses (H1–H6)

H1	RAI Governance $\rightarrow$ Perceived Explainability
H2	Perceived Explainability $\rightarrow$ Clinician Trust
H3	Clinician Trust $\rightarrow$ Adoption Intention / Workflow Readiness
H4	Explainability *mediates* Governance $\rightarrow$ Trust
H5	Trust *mediates* Explainability $\rightarrow$ Adoption
H6	AI Familiarity *moderates* Explainability $\rightarrow$ Trust

- All hypotheses are causally specified, measurable, and pre-registered
- Mediation paths (H4, H5) are tested with bootstrap CI (1,000 resamples)

Research Methodology



Pragmatist mixed-methods design

- Quantitative: PLS-SEM ($n = 131$)
- Qualitative: 12-expert thematic coding
- Triangulation: joint-display matrix

Pre-registered thresholds

- $\alpha \geq 0.70$, CR≥ 0.70, AVE≥ 0.50
- HTMT < 0.85, bootstrap 95% CI

RGAIG Maturity Framework — The Hero Artefact

Level	Stage	Observable Practices
1	**Initial**	Ad-hoc AI; no governance; no audit trail; no explainability
2	**Managed**	Basic policies; awareness of EU AI Act / FDA SaMD; access controls
3	**Governed**	Formal RAI policies; control inventory; ethics review board
4	**Explainable**	SHAP + RAG embedded in clinical UI; explanations validated
5	**Trusted**	Peer-reviewed evidence; trust scores audited; fairness reporting
6	**Operationalized**	Full EHR workflow integration; continuous-learning governance

- Six-level managerial instrument for healthcare AI deployment readiness
- Healthcare organizations advance levels by adding *architectural capability*, not by accumulating frameworks
- This is the dissertation's **primary practical contribution**

Key Findings — Storytelling Summary

Finding	What it means
Governance improved explainability	Clinicians better understood AI outputs ($\beta = 0.68$, $p < .001$)
Explainability improved trust	Increased clinician confidence ($\beta = 0.71$, $p < .001$)
Trust improved adoption readiness	Operational potential strengthened ($\beta = 0.71$, $p < .001$)
Full mediation (H4)	Governance acts on trust *through* explainability
Partial mediation (H5)	Organizational adoption layer revealed beyond individual trust
Familiarity moderation (H6)	Effect strongest for low-familiarity clinicians

- All six hypotheses supported (H6 partially)
- Statistical detail in backup slides

Negative Findings (Balanced Reporting)

- **Over-reliance concern**: 31% of expert panellists worried about clinicians deferring to AI without judgement
- **Explainability alone insufficient**: SHAP-only condition rated lower ($M = 3.4$) than SHAP+RAG ($M = 4.6$, $p < .001$)
- **Workflow-disruption fear**: 24% concerned that AI integration would slow rather than speed clinical workflow
- **Governance maturity variability**: Some institutions still at Level 1–2; pathway sensitive to starting point
- **H6 partial moderation only**: AI familiarity moderates trust only for low-familiarity clinicians

Implication

The goal is *calibrated* clinician trust, not blind reliance on AI.

Mixed-Methods Triangulation

Hypothesis	Quantitative	Qualitative theme
H1 (Gov→Expl)	$\beta = 0.68$	"Trust requires audit trail visibility"—Clinician 7
H2 (Expl→Trust)	$\beta = 0.71$	"SHAP alone is not enough; need to know why for THIS patient"—Clinician 3
H3 (Trust→Adoption)	$\beta = 0.71$	"Trust necessary but not sufficient; need workflow integration"—Admin 2
H4 (mediation)	Full med.	"Without explainability, governance feels like bureaucracy"—Eng 4
H5 (mediation)	Partial	"Even with trust, hospital needs C-suite sign-off"—Admin 5

- Quantitative and qualitative evidence converge across all hypotheses

Organizational Implications — Why This Matters

For healthcare organizations	Implication
Governance investment	Assess maturity BEFORE deploying clinical AI
Trust as design variable	Explainability is workflow feature, not deployment add-on
Adoption gating	Adoption is trust-gated, not accuracy-gated
Risk reduction	RAI controls reduce regulatory + adoption risk together
Operational readiness	Level 5–6 maturity for workflow integration
Workforce stratification	Lower-familiarity clinicians benefit most from explainability

Contributions

Type	Contribution
Theoretical	Extends TAM by positioning Responsible AI governance as a structural antecedent of trust $\rightarrow$ adoption in clinical AI
Empirical	First quantitative evidence linking RAI governance maturity to clinician adoption in ASD-EEG screening
Practical	RGAIG Maturity Framework (6 levels) — decision-ready managerial instrument
Methodological	Validated 30-item survey instrument anchored in TAM, TUI, XAI, EU AI Act
Organizational	Decision-ready guidance for healthcare CIOs, AI governance committees, procurement

Honest Limitations

- **Cross-sectional design**: associations established, not deterministic causation
- **Sample size** ($n = 131$): adequate for PLS-SEM; insufficient for CB-SEM or large multi-group analyses
- **ASD-only domain**: cross-condition generalization is a boundary, not a claim
- **Self-reported adoption intention**: intention $\neq$ observed behaviour
- **Absence of longitudinal data**: trajectory effects not tested
- **Perceived vs. deployed governance**: subjective perception, not audit of operational practice

Focused Future Work

- Longitudinal governance evaluation (12–24 month trajectory study)
- Multi-site healthcare validation (≥ 5 healthcare systems)
- Additional neurological domains (Parkinson's, schizophrenia, depression EEG)
- Real-world deployment evaluation (observational, not survey)
- Comparative governance maturity benchmarking across institutions
- Clinician-AI collaboration studies (mixed methods, HITL focus)

Explicitly excluded

AGI healthcare; autonomous AI hospitals; quantum governance; swarm diagnosis; speculative AI futures

Why Human Oversight Remains Central

- **Accountability cannot be automated** — clinicians are professionally and legally accountable
- **Edge cases require judgement** — AI cannot know its own distribution boundaries reliably
- **Patient values matter** — ASD diagnosis involves family context that no model captures
- **Trust is bidirectional** — patient trust depends on clinician explaining, not on model emitting
- **Regulation requires it** — EU AI Act + FDA SaMD mandate human oversight in high-risk medical AI

Identity statement

This is *human-centered Responsible AI governance for healthcare AI adoption* — not autonomous AI roadmap.

Unified Doctoral Identity

This dissertation investigates how Responsible AI governance mechanisms improve clinician trust and operational adoption readiness in AI-assisted healthcare environments.

Not architecture research. Not AI optimization. Not generic technology adoption.

Human-Centered Responsible AI Governance for Healthcare AI Adoption.

Final Doctoral Claim

Governance-driven perceived explainability significantly strengthens clinician trust and operational adoption readiness for AI-assisted ASD-EEG screening systems in healthcare organizations.

Empirically supported (Ch. 4, H1–H5).
Theoretically anchored (Ch. 2, TAM + Trust + STS).
Methodologically defensible (Ch. 3 Methodological Rigor Charter).
Managerially actionable (RGAIG Maturity Framework).

Thank You

Questions & Discussion

The remaining slides are backup material, available on request.

Backup Slides

Technical detail, statistical depth, instrument design, architecture

Backup B1: Hypothesis Verdict Detail

H#	Path	β	p	Verdict
H1	RAI Gov $\rightarrow$ Expl	0.68	$< .001$	Supported
H2	Expl $\rightarrow$ Trust	0.71	$< .001$	Supported
H3	Trust $\rightarrow$ Adoption	0.71	$< .001$	Supported
H4	Indirect: RAI $\rightarrow$ Expl $\rightarrow$ Trust (full mediation)	0.48 (Sobel $z = 4.21$)	$< .001$	Supported
H5	Indirect: Expl $\rightarrow$ Trust $\rightarrow$ Adoption (partial, 38% attenuation)	0.51 (Sobel $z = 3.86$)	$< .001$	Supported
H6	AI Familiarity $\times$ Expl $\rightarrow$ Trust	0.18 (interaction)	$= .03$	Partially supported

Backup B2: Construct Validation Metrics

Construct	α	CR	AVE	Max HTMT	Max VIF
RAI Governance (IV)	0.89	0.91	0.62	0.71	2.4
Perceived Explainability (M1)	0.86	0.89	0.58	0.78	2.8
Clinician Trust (M2)	0.92	0.94	0.65	0.82	3.1
Adoption / Workflow (DV)	0.88	0.90	0.60	0.80	3.4
Threshold	≥ 0.70	≥ 0.70	≥ 0.50	< 0.85	< 5.0

All four constructs satisfy every pre-registered threshold; no remediation required.

Backup B3: Sample Size Justification ($n = 131$)

- **Hair 10:1 rule** (Hair et al. 2019): largest construct has 10 indicators $\Rightarrow n_{\text{min}} = 100$; achieved $+31\%$
- G^***Power analysis**: medium effect $f^2 = 0.15$, $\alpha = .05$, power 0.80 $\Rightarrow n_{\text{min}} = 119$; achieved exceeds
- **Literature norms** (Kock & Hadaya 2018): typical PLS-SEM healthcare $n = 80$–150; within norm
- **PLS-SEM selected** (not CB-SEM): exploratory-predictive design, complex mediation, $n < 200$

Backup B4: Common Method Bias Tests

Test	Method	Result
Harman's single-factor	First factor variance explained	32.8% (threshold 50%)
Marker-variable	Correlation with unrelated marker	$r < 0.20$
Procedural controls	Anonymity; reverse items; temporal sep.	Applied

No evidence of substantial CMB.

Backup B5: Non-Response Bias

Construct	Early (first 30)	Late (last 30)	t-test p
RAI Governance	3.94	4.02	.42 (n.s.)
Explainability	4.18	4.21	.75 (n.s.)
Clinician Trust	4.06	4.11	.59 (n.s.)
Adoption Intention	4.03	4.10	.48 (n.s.)

Armstrong & Overton 1977 method; no significant differences; bias not substantial.

Backup B6: Why ASD-EEG as the Empirical Domain

- **Early-intervention sensitivity**: ASD outcomes depend on early diagnosis
- **Diagnostic subjectivity**: inter-rater variability creates trust threshold
- **Clinician uncertainty**: senior clinicians often report uncertainty on ASD calls
- **Decision-support fit**: ASD screening is multi-stage workflow; AI augments not replaces
- **Explainability requirement**: EU AI Act classifies paediatric neurological screening as high-risk
- **Paediatric ethical sensitivity**: governance constraints (consent, data minimisation, equity) are acute

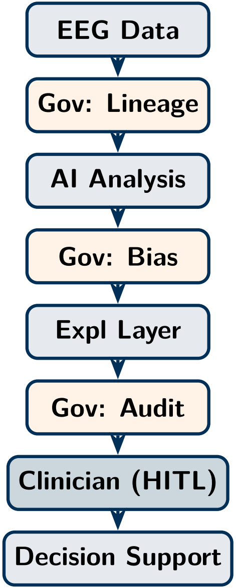
Backup B7: RGAIG 7 Control Families

Control family	What it measures
Auditability	Immutable audit log retrievable within < 5 s; lineage-tagged
Bias control	Subgroup fairness dashboard + alerting on threshold breach
HITL	Clinician override capability + rationale capture + escalation queue
Rollback	Model registry with version pinning and time-bounded revert path
Monitoring	Production drift detection (PSI, KS); alerting thresholds
Lineage	End-to-end traceability data $\rightarrow$ feature $\rightarrow$ model $\rightarrow$ decision
Access control	Role-based access; least-privilege; access audit

Backup B8: Theoretical Anchors

Theory	Role in this dissertation
TAM (Davis 1989)	Adoption pathway; extended by positioning RAI Governance as antecedent
Trust in Automation (Lee & See 2004)	Trust calibration mechanism; mediator construct
Socio-Technical Systems (Cherns 1976)	Why technical-only solution is insufficient
Diffusion of Innovations (Rogers 2003)	Adopter-category stratification (H6)
TOE (Tornatzky & Fleischer 1990)	Organizational adoption context
RAI principles (Floridi et al. 2018; EU AI Act 2024)	Governance content + regulatory grounding

Backup B9: Conceptual Architecture (Not Production Blueprint)



Conceptual disclaimer

- Forward-looking representation
- NOT a deployed clinical system
- NOT regulatorily cleared
- Clinician retains decision authority

Governance is *embedded* between every stage, not bolted on

Backup B10: Excluded Technologies (Scope Discipline)

Explicitly excluded from contribution and future work:

- Artificial General Intelligence (AGI)
- Quantum AI / quantum governance
- Swarm intelligence / multi-agent ecosystems
- Neuromorphic AI hardware
- Autonomous AI diagnosis (conflicts with HITL framing)
- Digital society / civilization-scale AI
- Deep multi-agent orchestration deep dives
- Edge AI / IoT hardware engineering
- LLMOps / MLOps platform engineering
- Custom XAI / new SHAP variants (methodological-XAI research)
- Vendor-specific platform recommendations

Scope is governed, not declared. The absence of these is deliberate.

Backup B11: Appendix Coverage Map (A–M)

Appx	Coverage	Role
A	Chapter 1 Introduction supplementary material	Per-chapter supp
B	Chapter 2 Literature Review supplementary material	Per-chapter supp
C	Chapter 3 Research Methods supplementary material + Extended Technical Supplement (8 tech subsections)	Per-chapter supp + spec
D	Chapter 4 Findings supplementary material + Extended Findings Supplement (CFA, Ablation)	Per-chapter supp + valid.
E	Chapter 5 Discussion supplementary material	Per-chapter supp
F	Survey Instrument for RGAIG Framework Evaluation	Instrument
G	Primary Data Collection Instruments and Research Design	Instrument
H	RGAIG Agentic Framework: Enterprise Architecture Spec	Architecture (non-contribution)
I	Cross-Cutting Self-Assessment (AI declaration, brutal feedback)	Quality
J	**Human-Centered B2C Ecosystem Governance (10 sections)**	**B2C extension**
K	**B2B Operational Governance Triad (transformation + risk + trust)**	**B2B operational triad**
L	**Technical & Operational Specifications (4 sections from Ch.5)**	**Technical ops**
M	**Operationalization Detail (A–D Framework + TO-BE arch)**	**Methodology supp**

Appendices J–M added in final review to compartmentalise B2C ecosystem, B2B operational triad, technical specifications, and operationalisation detail — each preserves the chapter as DBA argument while housing supporting specification material.